

CREDIT CARD TRANSACTION ANALYSIS AND FRAUD DETECTION SYSTEM

BY

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CHAPTER ONE:

INTRODUCTION

1.1 Background to the Study

Online financial services have experienced tremendous growth, and it has greatly changed the way financial transactions are made around the globe. In today's economies, the speed and volume of financial transactions have grown with the advent of digital banking platforms, electronic commerce systems and mobile payment applications. A financial transaction is the act of exchanging value between two or more parties in a monetary system either by electronic or physical means. These innovations have enhanced convenience, efficiency and financial inclusion, but they have also brought about new security issues in digital payment systems. It has been proved that a more interconnected financial system is more vulnerable to unauthorised access and fraudulent activities, especially within high-volume transaction networks (Boutaher et al., 2020; Pandey, Sachan, & Ganpatrao, 2021). This has rendered the need for continuous monitoring of transactions critical to ensuring the integrity and trust of electronic payment systems.

Credit cards are an integral part of today's electronic payments, particularly in e-commerce and digital banking. A credit card is a form of credit offered by financial institutions that can be used to obtain credit for goods or services up to a certain limit. Credit cards have become the natural extension of e-commerce, making it easier and more efficient to conduct transactions from a distance, but it has also raised concerns about security risks. The increased adoption of digital cards has been linked to an increase in fraud cases, mainly because of the anonymity and automation of online transactions (Bin Sulaiman, Schetinin, & Sant, 2022; Roseline et al., 2022). With the development of technology, fraud patterns are also changing, making it difficult for traditional security systems to be effective (Du et al., 2024).

Fraud with credit cards is still one of the greatest issues in the electronic financial systems. When someone uses a credit card to get something they did not ask for without the authorisation of the legitimate cardholder. Payment verification systems and behavioural tracking tools in financial platforms can be exploited by fraudulent activities, which can be found through the weaknesses in such systems (Sarkar et al., 2023; Gowhor, 2024). Some of the existing methods include rule-based filter systems and manual verification, which have been applied to identify suspicious transactions. These, however, are not practical for large

volumes and high-speed transaction processing where thousands of transactions can happen in seconds (Khanuja & Adane, 2020). As a result, researchers have been interested in automatic schemes that can more effectively detect unusual transaction behaviour.

In recent years, more and more research is directed at intelligent fraud detection systems, where web technologies are coupled with advanced data analysis techniques. Fraud detection systems are computational systems that can detect unusual patterns or anomalies in transaction information that could be a sign of fraud. In financial systems, a number of machine learning and real-time monitoring techniques have been suggested to enhance detection accuracy and response time (Hafez et al., 2025; Khalid et al., 2024). Due to factors like the complexity of system integration, data imbalance, and computational constraints, their comprehensive integration into real-world financial environments has been limited (Sun et al., 2025). Thus, the creation of a web-based system including transaction monitoring, fraud detection, and transaction analysis will continue to be crucial in enhancing the recognition and control of suspicious credit card transactions (Abutaleb et al., 2025; Hanae et al., 2023).

1.2 Motivation of the Study

Digital payment systems and online financial transactions have seen a lot of development in recent years, making financial transactions very convenient. But this progress has come at a cost of a rise in credit card fraud, causing significant losses to financial institutions, businesses and consumers. Fraudulent operations are becoming more complex, and there is a growing need for intelligent systems that can keep track of transactions and detect fraudulent activities in a timely manner.

This study's motivation comes from the desire to investigate the potential of modern Web technologies to be used to build a platform for fraud detection and transaction analysis. While commercial financial institutions utilise advanced fraud detection solutions, these are often proprietary and costly and not available for educational purposes. The prototype web-based fraud detection system is an opportunity to show how transaction monitoring, rule-based fraud detection and data analysis can be combined into one application.

In addition, the rise of secure electronic payment systems and cybersecurity in the financial sector are driving the motivation for this study. It is also a chance to learn and use some of the software engineering principles, web development technologies, database management and data analysis techniques to solve a real-life problem. The results of this research will be used

in academia to add value to the body of knowledge and to show a practical solution to support fraud detection and transaction analysis in a simulated environment.

1.3 Statement of the Problem

Financial transactions have become faster, easier and more accessible as the availability of electronic payment systems has increased in today's digital economies. Payment systems that are considered effective should include means for the secure processing of transactions and adequate means for monitoring activities that could be indicators of fraud. Transaction behaviours should be analysed to spot patterns that are not usual, suspicious activities and quick responses to potential financial risks. Past research has highlighted the need for sophisticated fraud detection tools, data analysis methods and automated surveillance for better fraud detection. Bin Sulaiman, Schetinin, and Sant (2022) pointed out the role of computational methods in the identification of credit card fraud, and Hafez et al. (2025) emphasised the need for scalable fraud detection architectures for handling large amounts of transaction data. Bin Sulaiman, Schetinin, and Sant (2022) mentioned the use of computational techniques to detect credit card fraud, and Abutaleb et al. (2025) stressed the significance of scalable fraud detection architectures and processing large volumes of transaction data. The studies show that to effectively detect frauds, there must be systematic transaction monitoring and analysis.

Nonetheless, it remains a challenge to accurately detect credit card fraud, even in light of these developments. The rising number and complexity of electronic transactions has led to the possibility of fraudulent transactions taking place in various transaction patterns. The traditional methods, which are based on manual verification, and the simple rule-based filtering process may not be sufficient to detect more complex and evolving fraudulent activity (Pandey, Sachan, & Ganpatrao, 2021; Shirodkar et al., 2020). Du et al. (2024) also pointed out false positives, that is, true transactions can be flagged as frauds, and false negatives, that is, fraudulent transactions may not be identified. These issues can make fraud monitoring less effective and make the transaction review process more cumbersome. Moreover, limited transaction analysis and visualisation may hinder the identification of relationships and patterns between transactions, by amount, location, frequency and other transaction properties (Sarkar et al., 2023; Khanuja & Adane, 2020). Therefore, the effectiveness of web-based transaction monitoring, fraud detection and transaction analysis needs to be further investigated.

Web and automated solutions to fraud detection and transaction monitoring have been proven to be relevant in recent studies. Hanae et al. (2023) explored real-time architectures for fraud detection in online transactions, and Khalid et al. (2024) explored computational techniques for enhancing credit card fraud detection. The problem of computational cost and complexity, detection performance, and analysis of transaction patterns are still reported (Sun et al., 2025; Hafez et al., 2025). In fact, there is a need to evaluate the capability of web-based fraud detection systems to assist with transaction monitoring, detection of suspicious transactions and transaction pattern analysis. This evaluation will be used to support evidence of the capabilities and limitations of web-based methods for credit card fraud detection and transaction analysis.

1.4 Aim and Objectives of the Study

Aim

- To design and implement a web-based system for credit card fraud detection and transaction analysis.

Objectives

1. examine the transaction monitoring capabilities of a web-based system for credit card transactions;
2. assess the effectiveness of rule-based and pattern-based techniques for identifying suspicious credit card transactions;
3. examine the transaction analysis capabilities of a web-based system for identifying abnormal transaction patterns; and
4. determine the usefulness of web-based transaction analysis in supporting the identification and investigation of suspicious credit card transactions.

1.5 Significance of the Study

The relevance of this study is not limited to financial institutions and payment and banking services companies; it is also relevant for cybersecurity players, software developers and financial technology researchers. These stakeholders are directly affected by the security, efficiency and reliability of electronic payment systems and thus have an interest in enhanced fraud detection and transaction monitoring capabilities.

The study provides a practical solution for financial institutions and payment service providers to better detect credit card transactions that are fraudulent. Institutions can utilise a web-based system that enables transaction monitoring, fraud detection, and analysis; this enables them to more effectively identify suspicious transactions and act faster to potentially limit the impact of financial threats. This can help to increase the security of transactions, minimise financial losses, and boost the confidence of users in digital payment systems.

The structured approach to detecting and analysing transaction behaviour could be useful to cybersecurity professionals and system administrators, as it offers them a more structured way to identify abnormal patterns in financial data. The study also provides software developers with a structure for creating and developing web-based fraud detection systems that may be modified or enhanced for implementation in the real world. The research adds to the existing body of knowledge for financial technology and cybersecurity researchers by illustrating how these three types of components transaction monitoring, rule-based detection, and analytical modules can be combined in a single web-based platform to provide enhanced fraud detection capabilities.

1.6 Scope of the Study

The paper is about the design and application of a web-based system that has been created to assist in credit card fraud and transaction analysis in electronic payment systems. The paper is focused on credit card transaction monitoring and analysis of the trends of the transactions with the view of detecting any suspicious behaviour that could be a sign of fraudulent behaviour. The system will be programmed to run in a web-based environment where the system users will be able to access the records of transactions and track the activities of transactions as well as make analyses of the patterns that can be linked to the possible cases of fraud. When testing and evaluating the system, past credit card transaction data is used to replicate a scenario of real transactions and determine the effectiveness of the fraud detection. The study will use simulated and historical transaction datasets for system development and testing. Live integration with banking systems will not be implemented. The scope thus includes the design of the system architecture, the integration of detection algorithms and the creation of analytical functionalities that aid in transactions, monitoring and detecting fraud in a web-based environment.

1.7 Limitations of the Study

There are some limitations which can affect the development and testing of the proposed system. One of the constraints has to do with limited access to actual financial transaction data because of privacy restrictions and protection of banking information. Consequently, the research can be based on artificial or externally obtained data which might not be representative of the intricacy of actual-world financial deals. The other weakness is that there might be a constraint in executing highly complex machine learning models due to the nature of computations, time constraints and the system development process. Moreover, development of the fraud detection system heavily relies on the quality, structure and size of the dataset employed in the testing and evaluation phases. Differences in the characteristics of the datasets can as such affect the effectiveness of the system in detecting fraudulent patterns of transactions.

1.8 Operational Definition of Key Terms

Credit Card Fraud: Unauthorized use of credit card information to conduct illegal financial transactions within the web-based fraud detection system developed in this study.

Fraud Detection: The process of identifying suspicious or unauthorized credit card transactions through analytical mechanisms implemented in the web-based system designed for this study.

Transaction Analysis: The examination of credit card transaction data within the developed web-based platform to identify unusual patterns that may indicate fraudulent activity.

Transaction Monitoring: The continuous observation and tracking of credit card transaction activities within the web-based fraud detection system implemented in this study.

Web-Based System: An online application developed in this study to monitor, analyze, and detect fraudulent credit card transactions through an internet-accessible platform.

CHAPTER TWO:

LITERATURE REVIEW

The review of related literature for this study is discussed under the following headings:

2.1 Conceptual Review

The conceptual framework of this study describes the relationship between the major variables that are part of the development of a web-based credit card fraud detection system. The framework is based on the interaction of web-based system components during transaction data processing and their impact on fraud detection. Web-based system components in this study are the independent variable. These elements consist of a system design, a transaction monitoring module, a fraud detection module, and a transaction analysis module. A web-based system is an application that is accessible via the internet and is built to track, process and report financial transaction activity on an integrated platform. Research has revealed that web-based fraud detection systems are capable of enhancing transaction monitoring and increasing the availability of fraud management processes in financial institutions (Adesola et al., 2024). Some researchers believe that web-based architectures offer advantages in terms of system flexibility and efficient monitoring, but others contend that web-based systems could raise integration and security issues in large financial networks (Ganesan, 2025). However, it's believed that a web-based system with both monitoring and analytical capabilities can enhance transaction management and fraud detection efforts.

Fraud detection performance is the variable (the dependent variable) in this study. Fraud detection performance is the capability of the developed system in correctly detecting fraudulent transactions in the system with the minimum number of false transaction classifications. Studies show that the performance of fraud detection systems is typically evaluated using various metrics, including detection accuracy, false positive rate, and false negative rate (Du et al., 2024). However, some researchers argue that AI-driven algorithms can achieve greater accuracy in detecting fraud than traditional rule-based systems, as the algorithms can analyse vast amounts of data and identify complex patterns in transactions (Hafez et al., 2025). But other research warns that such complex detection models might be hard to understand and own in real financial systems (Shirodkar et al., 2020). However, this position implies that the performance of fraud detection should not rely solely on the accuracy of detection but also on the practical ability of the system to process the transaction activities in a consistent manner in the web environment.

The mediating variable in the framework is transaction data processing. Transaction data processing refers to the gathering, structuring and analysis of the transactions before they are used for fraud detection. Research on transaction monitoring systems shows that the efficiency of transaction processing directly impacts the efficacy of fraud detection results, with poor or incomplete data affecting the confidence in detection (Sarkar et al., 2023). To enhance anomaly detection, researchers like Yu et al. (2023) stress the importance of transaction analysis systems' ability to handle both behavioural and transactional data, considering it from various angles. Other studies have concentrated primarily on the algorithmic performance with less consideration given to the quality of the transaction preparation and organisation prior to analysis (Nguyen et al., 2020). Thus, efficient transaction processing is still required to enhance the system components' relationship and reduce fraud detection inefficiencies.

The framework also acknowledges moderator variables which may affect the effectiveness of the proposed system. These variables are transaction data quality, system response time and transaction volume. Transaction data quality is based on the completeness, consistency and accuracy of the transactions the system is processing. A system's response time is how quickly transactions are detected and alerted for examination in light of a suspicion, and the transaction volume is the amount of transactions that are processed during a defined time interval. Poor quality data and high transaction volumes can negatively impact fraud detection systems, especially in real-time financial settings (Abutaleb et al., 2025). While some deep learning frameworks have been proposed to process large transactions, these methods typically require significant computational efforts which is not feasible for all applications (Roseline et al., 2022). Hence, it is suggested that the performance of fraud detection in this study will be influenced in addition to web based system components, by the way transaction data is processed in various operating conditions.

2.1.1 Web-Based Systems for Fraud Detection and Transaction Analysis

With the expansion of online financial services, e-commerce, and online banking, web-based financial systems have emerged as an integral component of the modern digital payment landscape. A web-based financial system is the financial system that is accessible via the internet and is used to process, monitor and control financial transactions in a centralised environment. It has been demonstrated that these systems have a positive impact on increasing access to financial services and speeding up the transaction process at different locations (Adesola et al., 2024). Others argue that web-based infrastructures offer flexibility

and enable ongoing monitoring of financial transactions in digital payment systems (Aschi et al., 2022). But other scholars state that internet-based environments are prone to security issues and threats to unauthorised access to financial systems (Shabbir et al., 2022). While the above arguments are valid, it is believed that properly designed web systems can serve as an effective platform to incorporate fraud monitoring and transaction analysis operations.

The architecture of Web-based transaction processing systems has been a topic of much discussion in recent research on fraud. In this context, system architecture is the structural organization of the software modules, databases, transaction interfaces and communication in a web platform. According to previous works, the successful transaction processing architecture should be able to collect data, validate transactions, monitor and report transactions in one system environment (Ganesan, 2025). In high-volume systems, the continuous transaction handling and scalability are amongst the advantages that make distributed architectures more favourable in the eyes of some researchers (Abutaleb et al., 2025). Other studies, however, attack the highly distributed systems for causing higher integration complexity and maintenance overhead, especially in systems that have limited computational resources (Sun et al., 2025). The critique indicates that the architecture design for fraud detection must be scalable, simple and easy to manage.

In addition, web platforms are also significant assets for providing real-time monitoring and access to transactions in financial systems. Real-time monitoring is a process that involves real-time observation and assessment of transactions that are taking place in the payment environment. Research has found that real-time monitoring systems lead to faster detection of suspicious activity and less time in fraud detection processes (Sarkar et al., 2023). Mohammad and Logeshwaran (2024) suggest introducing such analytical mechanisms into the web platforms so that suspicious transactions can be detected during the transactions, instead of after they are completed. However, some authors argue that real-time monitoring systems can be impacted by large volumes of transactions or poorly organised data (Khanuja et al, 2020). Therefore, it is proposed that a reliable monitoring performance should be achieved by efficient transaction processing mechanisms on the web platform in conjunction with its accessibility.

Fraud detection modules are now playing a crucial role in modern financial security systems, integrated into web applications. A fraud detection module is a part of a transaction system used to detect suspicious or unusual transaction activity that is defined by rules or analytical patterns. Research has shown that the addition of fraud detection systems to web systems can aid automated transaction analysis and lessen reliance on manual verification

processes (Njoku et al., 2024). Some argue that machine learning methods are suitable for fraud detection due to their ability to evolve and adjust to new fraud patterns and gain in detection accuracy over time (Hafez et al., 2025). However, other researchers claim that rule-based systems are still useful due to its easy interpretation and implementation in practical environments (Pundkar et al, 2023). However, it is believed that an integrated monitoring, analysis and fraud detection solution in a single centralized web-based application can streamline transaction management and facilitate automated identification of strange financial transactions.

2.1.2 Transaction Monitoring Systems Using Simulated and Real-Time Data

With the boom of electronic payment activities in digital environments, transaction monitoring has emerged as an important part of financial security systems. The continuous monitoring, recording and assessment of financial activities to detect unusual or suspicious activity as the transactions are processed is called transaction monitoring. Research shows that financial institutions employ monitoring mechanisms to monitor transactional patterns and mitigate the risk of unauthorised financial activities in electronic payment systems (Sarkar et al., 2023). Others believe that transaction monitoring systems need to facilitate the automatic analysis of suspicious transactions and prompt notifications to the fraud response team to speed up fraud response time (Aschi et al., 2022). However, some researchers argue that the success of monitoring systems relies heavily on the quality of transaction information and the processing techniques used in the design of monitoring systems (Khanuja et al, 2020). The position presented here is that transaction monitoring needs to be more than just detection technology and instead involve management and organisation of transaction information.

There has been much research on real-time versus simulated transaction data in the design and analysis of fraud detection systems. Real-time transaction data is data from transactions that are processed in real time while financial transactions are happening in a real-time payment system environment. Simulated transaction data, however, refers to either historical or generated data sets that are employed to simulate real transaction activities for the purpose of testing and developing. The researchers have pointed out that the real-time monitoring systems are faster to detect frauds, as they can be identified during the transaction process instead of after its completion (Hanae et al., 2023). However, there are other studies that suggest that using simulated data within experimental settings is an acceptable approach, as it offers a controlled environment to assess the performance of the system without

compromising financial data (Adesola et al., 2024). While real-time systems offer practical monitoring tools, using these tools in academic research settings is challenging due to the lack of access to live banking infrastructures and transactions.

Various approaches have been used to monitor finance transactions in fraud detection systems. Previous research has established that some common methods for observing financial activities in the electronic payment environment are transaction logging, behavioural tracking, anomaly monitoring and database recording (Shabbir et al., 2022). Transaction logging is the process of making a record of what is happening within a transaction to review and analyse later. Some scholars like Yu et al. (2023) believe that transaction tracking systems should use several behavioural features, such as transaction amount, location, frequency and spending pattern, to better identify the anomalies. Researchers, however, have concerns about systems that heavily depend on single pattern analysis for their failure to detect the new fraud behaviours in a complex transaction environment (Bronskaya et al., 2024). Therefore, it is concluded that the effective transaction monitoring system should consist of a combination of continuous recording and behavioural analysis, which helps in the proper identification of fraud.

Historical data is also played an increasingly significant role in research for testing and simulation of fraud detection systems. Researches have shown that simulated transaction datasets provide researchers the opportunity to gain insights into the behaviour of the system, determine the accuracy of detection, and test analytical models under controlled conditions without needing to access financial institutions (Boutaher et al., 2020). However, there are some scholars who believe that historical data sets may not adequately reflect the evolving characteristics of fraud or the live transaction context (Sun et al., 2025). Also, the deployment of fraud detection systems in academia is still not ubiquitous due to limited access to banking infrastructures, privacy policies, and computational needs for real-time transaction processing (Roseline et al., 2022). Thus, simulated data are considered as a practical solution in academic research contexts for the evaluation of the transaction monitoring and fraud detection processes when used in web-based systems.

2.1.3 Rule-Based and Pattern-Based Fraud Detection Techniques

Many credit card fraud detection research studies still discuss rule-based and pattern-based methods. A rule-based fraud detection system is a system that monitors for fraud by applying a set of rules or conditions set by system administrators. These rules could be based on transaction amounts, an unusual number of transactions, or an unusual location within the

payment environment. Research shows that rule-based systems are widely adopted due to their ease of implementation and because they can be directly interpreted to detect suspicious transaction behaviour (Pandey et al. 2021). Some researchers claim that this is the case in known fraud patterns in structured financial environments (Adesola et al., 2024). This is, however, where other scholars question rule-based approaches, citing the need for the need to take into account evolving fraud tactics over time (Shirodkar et al., 2020). Rule-based techniques are thus still relevant for basic fraud detection and also need analytical assistance in order to achieve a wider impact.

The engine of a rule-based fraud detection system is the use of predefined conditions. Predefined conditions are the specific conditions for a transaction that are pre-defined and used to determine whether the transaction is suspicious or not. Financial institutions have been found to use rules such as transaction amount, transaction time, transaction frequency and transaction location to detect irregular financial transactions (Sarkar et al., 2023). Others believe that predefined rules enhance transparency because there is an easy explanation why a transaction was flagged by the system (Khanuja et al, 2020). However, other research has noted that relying too heavily on strict rules can lead to false positives, meaning that legitimate transactions may be flagged as fraudulent (Du et al., 2024). However, if the fraudsters change their tactics to avoid detection conditions, some of these transactions will not be detected. This view implies that a fraud detection system should not rely exclusively on rules for detecting fraud in a changing transaction environment.

The pattern-based fraud detection techniques have been developed as a viable alternative for detecting suspicious financial activities by analysing behavior. Pattern-based detection is the process of analysing historical transaction behaviour to detect anything that is unusual in terms of spending activities. Anomaly detection in financial data relies on existing studies that have focused on patterns of spending, transaction sequences, purchase locations, and transaction timings to identify anomalies in financial data (Boutaher et al., 2020). Analytical models that are based on behavioural patterns provide higher accuracy of fraud detection because they change their models over time in response to changes in transaction activity in the system (Hafez et al. 2025). Despite their advantages, however, pattern-based approaches are sometimes complex and require extensive data volumes and computing power to be properly applied (Sun et al., 2025). This critique points to the feasibility of implementing very sophisticated analysis systems in a place where infrastructure support is limited.

Identifying abnormal spending habits continues to be key to pattern-based fraud detection systems. An anomaly is an action that is unusual or inconsistent with a transaction that is expected based on a known spending trend for a cardholder or financial account. Previous studies also demonstrated that anomaly detection techniques are applied to detect anomalous transaction amounts, anomalous purchase locations, or anomalous rapid change in spending patterns from transaction data (Nguyen et al., 2020). Others argue that the incorporation of machine learning models is warranted as they provide the ability to acquire new knowledge about fraud and adapt to new fraud trends in the constantly changing financial landscape (Roseline et al., 2022). Other research, however, has challenged the ease of interpretation and control of highly adaptive systems in real-world fraud investigations (Bin Sulaiman et al., 2022). Therefore, it is believed that a combination of rule-based simplicity with behavioural pattern analysis could lead to a balanced way of enhancing fraud detection effectiveness in web-based transaction systems.

2.1.4 Transaction Analysis and Behavioural Pattern Detection Systems

Due to the fact that electronic payment activities are becoming more and more complex, transaction analysis has emerged as an important tool for financial fraud management. Transaction data analysis involves the analysis of financial transaction data to detect trends, irregularities, and suspicious activity in payment systems. Previous research has shown that financial institutions are using transaction analysis systems for tracking customers' behaviors and enhancing the process of fraud detection (Aschi et al., 2022). Some researchers suggest that transaction analysis should be limited to the transaction amount and the number of transactions as these are measurable evidences of abnormal transactions (Pandey et al., 2021). However, it is argued that analysing transaction data alone can result in misdetections if the context of the user's behavior is not taken into account (Yu et al., 2023). The criticism indicates that a transaction analysis system should combine behavioural data with transactional data to enhance fraud detection in digital payment systems.

Regular credit card usage patterns are often used to verify against current credit card transactions with behaviour pattern detection systems. Behavioural patterns are defined as repeated spending habits relating to a specific cardholder, such as the location of the transaction, amount, transaction timing and spending frequency. Studies have revealed that the ability of detecting normal transaction behaviour enables fraud detection systems to better detect unusual financial transactions (Nguyen et al., 2020). Other scholars suggest using adaptive analytical models since they may learn transaction behaviour with time and adapt to

variations in spending activities (Hafez et al., 2025). However, other research has warned that such systems could misidentify atypical but valid transactions as fraudulent, particularly in the event of sudden changes in spending patterns (Du et al., 2024). Therefore, it is argued that transaction verification tools should be used along with behavioural analysis to minimise false transactions in fraud detection systems.

Detection of anomalies, based on normal and unusual transaction behaviour, is a key aspect of behavioural fraud analysis. An anomaly is an activity on a transaction that is substantially unlike a set behavioural pattern in a financial system. Research shows that anomaly detection systems compare the activity of the current transactions against the historical transactions to detect anomalies related to potential fraud (Boutaher et al., 2020). On the other hand, some researchers state that the anomaly detection method can also help to identify frauds that may not be identified using predefined rules, thereby increasing the effectiveness of fraud identification (Khalid et al., 2024). However, others have pointed out that high-volume transaction systems may generate too many alerts for anomaly-based systems, placing a burden on fraud analysts (Shirodkar et al., 2020). The recommendation includes the integration of filtering and prioritisation features into the anomaly detection systems to enhance the efficiency of monitoring and reduce the number of false alarms.

Transaction analysis systems are also important in grouping the data and analyzing trends. Data grouping is the process of grouping transaction records based on the common behaviour that they share, and trend analysis is the process of detecting the transaction behaviours that repeat over a defined time frame. Some researches have been conducted to show that the clustered records of transactions facilitate fraud investigation as they allow comparing suspicious activities across different behavioural groups and transaction histories (Bronskaya et al., 2024). Visualization further enhances transaction analysis, as illustrated by Abutaleb et al. (2025) who contend that visualization tools would present transaction activities in the form of charts, tables, and graphical interfaces, which would aid in the interpretation of the behavioral trends. But there are certain studies which note that the analysis of large and complex transaction data is still challenging due to the growing number of transactions, the existence of imbalance in data, and the need to apply computational resources to real-time monitoring systems (Sun et al., 2025). Thereupon, it is assumed that effective transaction analysis systems need to integrate behavioral detection, structured data visualization and presentation, and analytical interpretation to enhance fraud monitoring and analysis in financial systems over the Web.

2.2 Review of Related Works

Adesola, Christiana, Aduragba and Adeyinka (2024) designed and implemented a web-based credit card fraud detection system for the 2024 International Conference on Science, Engineering and Business for Driving Sustainable Development Goals (SEB4SDG), which aims to detect and analyse credit card fraud in real-time. The study was relevant to the current research, as it was conducted in a web application environment that is similar to the intended application of the system and integrated fraud detection mechanisms. The goal of the study was to create a web-based system that can identify fraudulent credit card transactions based on the structured data in transactions. A descriptive system development research design was used, which involved building a prototype system and evaluation. There was no actual number of records used as a sample, as there were no humans used as participants in the simulation of transaction data. The study took place in a software development controlled environment and not in a financial institution. Data collection involved using simulated credit card logs in tabular format, and the main data source was the system logs. Rule-based detection logic and simple statistical assessment of the performance of the systems were used for analysis. It was found that web-based systems provide better accessibility and enable centralised monitoring of transactions but suffer from some shortcomings when it comes to tackling complex fraud scenarios. The research found that web integration will improve fraud detection efficiency, but analytical modules need to be improved to increase accuracy. Both studies have the goal of creating a fraud detection system that runs on the web and uses simulated transaction data. But the reviewed study was mainly concentrating on basic rule-based detection, while the current study integrates transaction monitoring, analysis and pattern detection modules in a more structured manner. The identified gap is the lack of integration of behavioural analysis and real-time-like monitoring, which is addressed by the multi-module system design in the present study.

Mohammad and Logeshwaran (2024), in their study "Real-Time Credit Card Fraud Detection using Deep Learning-Based Framework", presented at the 2024 International Conference on Advances in Computation, Communication and Information Technology, proposed development of a deep learning framework for real-time detection of credit card fraud. The study was relevant to the current research, as it spoke about the real-time fraud detection using advanced analytical techniques. The goal was to create a deep learning model that can detect fraudulent transactions with a high degree of accuracy. The approach to the design of this experimental research was by means of machine learning models being trained

and tested with historical credit card data. The data set employed was public transactional records, though the number of the records in the data set was not specified. The study setting was a computational simulation setting. The data used for the construction of the structure was collected from structured transaction datasets, and feature extraction methods were used to prepare data for the model. Neural network algorithms are used for analysis, and the evaluation was done using the performance metrics: accuracy, precision, and recall. Results showed that deep learning models outperformed traditional machine learning techniques in terms of accuracy when detecting fraud, but they also were more complex. The study found that deep learning is able to improve the fraud detection performance, but it also demands high processing power. Both the reviewed and current studies focus on credit card fraud detection, but the reviewed study highlights deep learning and real-time prediction techniques, whereas the current study uses the rule-based and pattern-based techniques in a web-based analytical system. One of the weaknesses found was the reliance on high computing resources, restricting the deployment to simple web environments. This gap is the justification for a more lightweight and integrated system, as proposed in the current research.

Roseline, Naidu, Pandi, Rajasree and Mageswari (2022) proposed an autonomous credit card fraud detection system using a machine learning approach in the study titled “Autonomous Credit Card Fraud Detection Using Machine Learning Approach” in "Computers and Electrical Engineering". The study was relevant as it covered automated fraud detection techniques used in credit card systems. The goal was to create a machine learning-powered solution that could automatically detect fraudulent transactions. A quantitative experimental design was employed, with training of classification algorithms using historical transaction datasets. The data was secondary and anonymised credit card transaction data. The study was carried out in a computational environment which was simulated. The data consisted of structured data provided with labelled transactions of fraud and non-fraud. Supervised learning algorithms and metrics such as accuracy and F1-score were used for analysis. The results revealed that machine learning can greatly enhance the efficiency of fraud detection, but it also faced challenges due to data imbalance. The researchers found that autonomous systems could be more efficient at detecting objects but needed balanced data sets in order to achieve this. Compared with the current study, both studies are related to automated fraud detection, but the reviewed study uses machine learning classification models, while the current study is concentrating on using rule-based and pattern-based techniques to detect frauds and integrates these techniques into a web system. The gap identified is in terms of the imbalance of data and limited interpretability of

the system, and the current study tackles this issue by using structured pattern analysis and rule-based logic.

Sarkar, Lincy, Sasireka and Mittal (2023) developed a transaction monitoring system in their study titled “Monitor and Detect Suspicious Online Transactions” in the International Journal of Electronic Security and Digital Forensics that aims to detect suspicious online financial transactions. The study was relevant since it was about transaction monitoring and behavioural analysis in digital environments. The goal was to create a system that will continuously monitor transactions on the Internet and identify unusual user activities. The system design approach used was descriptive, and the transaction monitoring logic was implemented and tested with the historical data sets. The synthetic logs used were a sample of transaction logs created for the purposes of the tests. The study was carried out in a controlled laboratory setting. Data collection was done by simulated transaction generation tools, and monitoring outputs were recorded for analysis. Analytical methods were used such as statistical comparison and anomaly detection algorithms. Results showed that continuous monitoring resulted in better detection of suspicious transactions but that there were scalability issues. The study found that monitoring systems are successful for fraud identification but need better data processing skills.

Sun, Shen, Gao, Mao, Qi, and Xu (2025) present an advanced fraud detection model in their study, “Fraud Detection Under Extreme Imbalance in Bank Account Opening”, which tackles extreme data imbalance in financial systems through computation. The study was relevant since the solution it provided was about a major problem in fraud detection systems, class imbalance in transactional datasets. The goal was to use optimised detection techniques to enhance the performance of fraud detection in a scenario with highly imbalanced data. An experimental research design was adopted wherein computational models were tested using large-scale financial datasets. The records of opening a bank account had extremely skewed class distributions. The study was carried out in a simulated environment for high-performance computing. The data collection consisted of structured financial data, and imbalance was addressed by using preprocessing techniques. Optimisation algorithms and classification performance measures were used to perform the analysis. The results showed that the imbalance has a significant impact on the accuracy of fraud detection, and optimised models can increase the accuracy of fraud detection. The researchers found that solving the data imbalance problem is essential to enhance the fraud detection system. Overall, both the reviewed and the current studies concern fraud detection problems, and the former one caters to optimisation in the presence of data imbalance, whereas the latter one concentrates on

system design and integration of the monitoring, detection and analysis modules in the web-based context. The gap identified is the lack of integration at a system level, which is solved through a single web-based fraud detection system in the present study.

2.3 Summary of Literature Review

In this chapter, the literature reviewed shows a fair amount of effort has been taken to tackle the problem of credit card fraud using web technologies, machine learning algorithms, deep learning models, and artificial intelligence techniques. Previous research has made many contributions, including the development of frameworks, architectures and predictive models to better detect fraudulent transactions accurately and efficiently. Analogously, a number of researchers have worked on transaction monitoring systems that enable ongoing monitoring of financial transactions and aid in fraud detection.

Even with these efforts, the reviewed research studies show some limitations. Studies focused mostly on improving the accuracy of fraud detection and did not consider transaction analysis, visualisation and reporting processes that are user-friendly. Some other studies were theoretical reviews or architectural frameworks that did not have functional web-based applications. Furthermore, most of the machine learning and deep learning methods demand huge amounts of labelled data, significant computational power, and complicated implementation procedures, which are unsuitable for lightweight educational or prototype systems.

In addition, enterprise-orientated fraud detection architectures suggested in some research works add implementation complexities that cannot be applied to the small-scale web applications. These constraints suggest that there is a requirement for a streamlined system that will be web-based and can monitor transactions, apply a rules-based fraud detection mechanism, alert the fraud, and provide the user with a user-friendly way to present the transaction analysis.

The findings from the reviewed literature are summarized in Table 2.1.

Table 2.1: Meta-Analysis of Related Literature

Author	Strength	Weakness	Limitation
Adesola et al. (2024)	Developed a web-based platform for credit card fraud detection with an intuitive user interface and effective fraud identification features.	Provided limited support for transaction analysis and visualization of transaction trends.	Focused primarily on fraud detection without integrating comprehensive transaction monitoring and analytical

			reporting.
Bin Sulaiman, Schetinin and Sant (2022)	Presented a comprehensive review of machine learning techniques used for credit card fraud detection and highlighted their comparative performance.	Did not include the design or implementation of a practical fraud detection system.	The study remained theoretical and offered no deployable solution for real-world application.
Khalid et al. (2024)	Improved fraud detection performance through ensemble machine learning models with high predictive accuracy.	The proposed approach required significant computational resources for model training and execution.	Dependence on large, labelled datasets limits its suitability for lightweight web-based applications and educational prototypes.
Mohammad and Logeshwaran (2024)	Proposed a real-time deep learning framework capable of accurately detecting fraudulent credit card transactions.	The framework involved complex implementation and extensive computational requirements.	The proposed solution is difficult to deploy in low-resource environments due to its reliance on deep learning infrastructure.
Hafez et al. (2025)	Provided a comprehensive review of recent artificial intelligence techniques for credit card fraud detection.	Emphasized AI algorithms while giving limited attention to practical system implementation.	Did not develop or evaluate a functional web-based fraud detection application.
Hanae et al. (2023)	Designed an end-to-end architecture for real-time fraud detection with clearly defined system components.	Focused mainly on architectural design rather than end-user interaction and usability.	Offered limited discussion on transaction visualization, reporting, and user-centered analytical features.
Njoku et al. (2024)	Developed a web-based fraud detection application with effective prediction capabilities using machine learning techniques.	Placed greater emphasis on prediction accuracy than transaction analysis and reporting.	Lacked comprehensive dashboards and analytical tools for monitoring transaction behaviour over time.
Abutaleb et al.	Proposed an optimized	The architecture was	Its complexity and

(2025)	architecture for fraud detection in large-scale big data environments with enhanced processing efficiency.	designed primarily for enterprise-level deployment, increasing implementation complexity.	infrastructure requirements make it less appropriate for academic prototypes and small-scale web applications.
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2.4 Research Gap

The literature review suggests that although much work has been done on the development of credit card fraud detection systems, there are still some gaps that have not been addressed. Comparatively few studies have concentrated on offering a full transaction analysis, monitoring and visualisation within a single web-based application, including the improvement of the accuracy of fraud detection, which is usually done using machine learning, deep learning or artificial intelligence methods.

Furthermore, numerous of the reviewed systems are based on sophisticated computational models which are highly demanding in their training sets and require high computing power and thus are not suitable for lightweight educational or prototype settings. Other studies are theoretical or architectural in scope, without offering practical implementations that show how transaction monitoring, fraud detection and analytical reporting can be integrated into practice.

To overcome these identified gaps, a web-based credit card fraud detection and transaction analysis system, which combines transaction monitoring, rule-based fraud detection, fraud alert generation and transaction analysis in a single platform, is designed and implemented. The proposed system is designed to detect and analyse frauds using simulated transaction data, unlike many previous studies which rely on large datasets and computationally intensive machine learning algorithms. As a result, the study offers an easy-to-use, user-friendly and cost-effective solution, applicable to academic research, learning environments and prototyping for the development of web-based financial security systems, and adds to the existing body of knowledge.

CHAPTER THREE

SYSTEM ANALYSIS AND DESIGN METHODOLOGY

This chapter will present the methodology that will be adopted for the design and implementation of the proposed web-based credit card fraud detection and transaction analysis system. The theoretical foundation that will guide the study will first be discussed, followed by the conceptual framework illustrating the proposed solution. The software development methodology, requirement engineering process, system design, architectural design, database design, computational framework, and implementation tools that will be employed during system development will also be described. Collectively, these methodologies and techniques will provide a structured approach through which the objectives of the study will be achieved and the proposed system will be implemented effectively.

3.1 Conceptual Framework

The concept of this study will be set up on the basis of the interaction of web-based transaction management, transaction monitoring, behavioural analysis, fraud detection rules, and making an administrative decision. The system being proposed will be designed in a way that will ensure that all transactions which are entered by users will be monitored, analysed and classified based on the fraud detection criteria defined beforehand before they are completed.

A web-based platform will be used, as it will offer access, scalability and central control over transactions. Adesola et al. (2024) state that these fraud detection systems, which are implemented on the internet, will enable efficient processing of financial transactions and assist in the automatic monitoring of fraud. In the same way, Prusti et al. (2021) report that web-based transaction processing systems will enable “continuous” transaction verification using built-in fraud detection systems.

The proposed system's main operational process will be the transaction monitoring. All transactions will be reviewed right after they are entered to detect any unusual transaction attributes that could alert them to fraudulent activity. Hanae et al. (2023) said that real-time monitoring architectures would greatly enhance the detection of frauds, as suspicious activities would be identified before the time of the transaction. Similarly, Abutaleb et al. (2025) suggested an optimised architecture that features simultaneous use of transaction monitoring, data processing and fraud analysis to enhance detection efficiency.

Support for the fraud detection process will be provided via predefined rules, as well as behavioural pattern analysis. Specifics of those transactions will be compared to thresholds that represent the nature of the transaction, frequency of transactions, location where they were logged in, timing of transactions, and the type of device they used. This was also mentioned by Sarkar et al. (2023), where they said they will be continuously monitoring the behavioural patterns and characteristics of transactions to identify suspicious ones.

The conceptual framework will also include an administrative review process, in which transactions that appear suspicious will automatically be reviewed. When the database forensic analysis is added to the fraud detection workflow, along with the administrative verification, the suspicious transaction monitoring systems will be strengthened, according to Khanuja and Adane (2020).

The proposed conceptual framework will therefore show how user transactions will go through authentication, transaction processing, fraud analysis, alert generation, administrative review and reporting. The relationship between these elements will ensure that fraudulent actions will be identified as soon as possible and that legitimate transactions will be carried out without unnecessary slowdown.

3.2 Theoretical Framework

Theories related to fraud detection, anomaly detection, behavioural analysis and transaction monitoring will be the bases for the proposed study. The various theories will serve as the basis for the proposed fraud detection mechanism.

i. Rule-Based Fraud Detection Theory

The first theory that will be the backbone of the proposed system will be the rule-based fraud detection theory. In this theory fraudulent activity will be detected whenever there's a violation of predefined rules in the transaction processing. The rules for the amount of transactions, number of transactions, login location, authentication failure, and timing of transactions will be decided before implementing the system.

Rule-based fraud detection is one of the most sure methods for financial transaction security, as suspicious activities will be detected right after the conditions are violated, as discussed by Aschi et al. (2022). Likewise, Ganesan (2025) states that the architectures of enterprise fraud detection systems shall continue to use rule-based systems due to their transparency, simplicity, and ease of implementation.

In the proposed system, all the transactions will then be checked against the set rules for fraud detection prior to the approval process.

ii. Anomaly Detection Theory

The anomaly detection theory also will be used as a theoretical base for the proposed system. According to this theory, fraudulent transactions will be identified as variations from the pattern of "legitimate" user transactions.

Financial transactions can be detected as fraudulent using anomaly detection methods which will be able to identify financial transaction characteristics that deviate from the historical financial transaction behaviour (Du et al., 2024). Likewise, Mohammad and Logeshwaran (2024) described how the real-time anomaly detection frameworks would enhance the effectiveness of fraud detection by continuously monitoring transaction activities.

The proposed system will then compare the behaviour of the transactions with the expected behaviour and see if there is any suspicious characteristic that will appear.

iii. Theory of Behavioural Pattern Analysis

The proposed fraud detection mechanism will also be assisted by the Behavioural Pattern Analysis Theory. According to this theory, fraud attempts will be identified as they are made, based on the ongoing monitoring of user behaviour patterns, not just the amount of the specific transaction.

YU et al. (2023) found that behavioural analysis methodologies would help in enhancing fraud detection accuracy by analysing user interaction patterns, transaction sequences, geographical locations of the users, and transaction frequencies. Likewise, Gowhor (2024) reiterated that integration of the behavioural indicators into the fraud detection system will make suspicious transaction reporting systems more effective.

Therefore, during the use of the system, behavioural information will be analysed constantly to detect deviation from existing user behaviour.

iv. Transaction Monitoring Theory

The Transaction Monitoring Theory will be used to understand why financial transactions should be constantly monitored throughout their lifecycle. According to this theory, transaction data will be reviewed on the spot after it is created, thus allowing for immediate identification of fraudulent activity and preventing any loss.

Shabbir et al. (2022) state that the continuous transaction monitoring will make financial security more robust by notifying about suspicious banking transactions before they are completed. Similarly, Hafez et al. (2025) found that fraud detection systems of the future will

be more sophisticated and rely on continuous monitoring and AI-based decision-making processes to enhance their accuracy of fraud detection.

The suggested system will therefore constantly monitor each transaction so that suspicious activities may be automatically detected and then the necessary administrative measures may be taken.

3.2.1 Development Methodology (SDLC)

For the development of the proposed web-based credit card fraud detection and transaction analysis system, the Iterative Waterfall Software Development Life Cycle (SDLC) model will be used. The model will be used, as it has a systematic and well-structured approach to software development, and it allows improvements to be added as needed throughout the development process. The outputs from each phase of the model will provide inputs for the next phase of the model, allowing the project to meet the project objectives in an organised and logical way.

The iterative waterfall model will be used in the proposed study, as the project requirements will be well-known from the beginning of the project. Furthermore, a relatively limited size of the system will allow for the application to be appropriate in that it will make it easy to document, test, and progressively refine the application before deployment. In contrast to a purely linear development, the iterative process will allow for changes to be made at any time when gaps are discovered while implementing the development or during testing.

The process of development will begin with the requirements analysis phase, where the functional and non-functional requirements of the proposed system will be identified and documented. This phase will involve development of the functionalities and performance characteristics of the app that are expected prior to its development.

This second phase will be the design of the system. In this stage, the general architecture of the application will be designed, the Unified Modelling Language (UML) diagrams will be created, the database structure will be elaborated, and the process models will be developed. These design artefacts will provide a blueprint that will guide the implementation of the proposed system.

Once the design step is completed, the implementation step will be followed. In the implementation stage, the designed modules will be converted to executable code using the selected programming languages, development frameworks, and database technologies. The

individual modules like transaction monitoring, fraud detection, transaction analysis, etc. will be built and incorporated to create the entire web application.

The system will then be tested to validate that it meets the requirements. Functional testing will be conducted to validate each module's functionality, and integration testing will be conducted to validate the correct communication between all the system components. If there are any discovered defects, they will be fixed prior to deployment.

The deployment phase will be performed after the successful completion of the testing phase. The finished application will be installed in the desired operating environment during this phase and set up for operation. Lastly, the maintenance phase will allow for future improvements, software updates and bug fixes, when needed, to keep the system running smoothly and dependably.

For this reason, the iterative waterfall approach will be the most suitable approach to the design and development of the proposed system, which is to be tested and maintained effectively.

3.2.2 Requirement Engineering

Requirement engineering will lead to systematic identification, analysis, documentation and validation of the requirements on which the proposed system will be built. In this way, before implementation is started, the functions expected from the system and the quality properties that need to be fulfilled in order to make the system function will be determined. Good requirement engineering will minimise ambiguities when coding and will also help secure that the final product will meet the goals set out in the requirements.

It will be derived from the review of relevant literature, existing fraud detection systems and functional needs of credit card transaction monitoring applications. Primary data collection in the form of questionnaires or interviews will not be necessary as the proposed work will be a software prototype and not the development of a commercial banking application. Rather, the system requirements will be identified on the basis of the practices followed in the field, in scholarly literature and in specific needs of the present study.

The identified requirements will be separated into functional and non-functional requirements. Functional requirements will be concerned with what services the proposed application will offer, and non-functional requirements will encompass quality attributes that the system will be expected to demonstrate while it is running.

3.2.2.1 *Functional Requirements*

The functional requirements will outline the minimal functions that the system will be able to carry out to meet the goal(s) of the study. These requirements will shape the action of the application from the authorised user's point of view and will have an impact on the processing, monitoring, analysis and presentation of transaction information.

The proposed system will be capable of securing the user's access, allowing only authorised users to access the application. Credit card transactions will be simulated and tracked on the fly as they are created in the system. Each transaction will then be reviewed with the predefined fraud detection rules to check if there is any suspicious activity.

If the user is alerted to any suspicious transactions, a fraud alert will appear automatically and be reviewed by the user. The application will also store a detailed history of the transactions, allowing users to search for specific transactions, view their account history, and trend transactions over time. In addition, graphical reports and analytical summaries will be created to help detect unusual transaction patterns and to make informed decisions.

Table 3.1: Functional Requirements

S/N	Requirement
1	The system shall provide user authentication.
2	The system shall record credit card transactions.
3	The system shall monitor transactions continuously.
4	The system shall apply fraud detection rules to each transaction.
5	The system shall generate fraud alerts for suspicious transactions.
6	The system shall maintain and display transaction history.
7	The system shall provide transaction search functionality.
8	The system shall generate transaction analysis reports and dashboard visualisations.

3.2.2.2 *Non-Functional Requirements*

Non-functional requirements will outline the quality attributes that the system will be required to exhibit in operation. Functional behaviour is not directly specified, but these requirements are likely to impact the reliability, usability, performance and security of the application.

The proposed system will be designed to give an acceptable performance level, processing the simulated transactions quickly and efficiently, with a quick response time. Suitable authentication methods will be integrated to improve the security of the system and

prevent any unauthorised access to transaction information. The user interface will be user-friendly by facilitating the navigation and interaction. Additionally, the application will be designed to allow for easy future maintenance and expansion when needed for additional functions to be added.

Table 3.2: Non-Functional Requirements

S/N	Requirement
1	The system shall provide secure user authentication.
2	The system shall process transactions within 3 seconds under normal operating conditions.
3	The system shall automatically terminate inactive user sessions after 15 minutes.
4	The system shall maintain reliable operation during normal usage.
5	The system shall provide high availability throughout its operation.
6	The system shall provide a simple, consistent and easy-to-navigate user interface.
7	The system shall be scalable to accommodate future enhancements.
8	The system shall be maintainable through modular software design.

3.2.3 System Design (UML)

The proposed web-based financial transaction monitoring and fraud detection system will be designed with Unified Modelling Language (UML) diagrams to visually represent the structure, interactions with users, workflow and relationships between objects of the system. The UML diagrams shall be used to gain better understanding of the proposed system before implementation and to ensure that all functional requirements shall be adequately represented. The following UML diagrams will be presented: Use Case Diagram, Class Diagram, Activity Diagram and Sequence Diagram.

Use Case Diagram

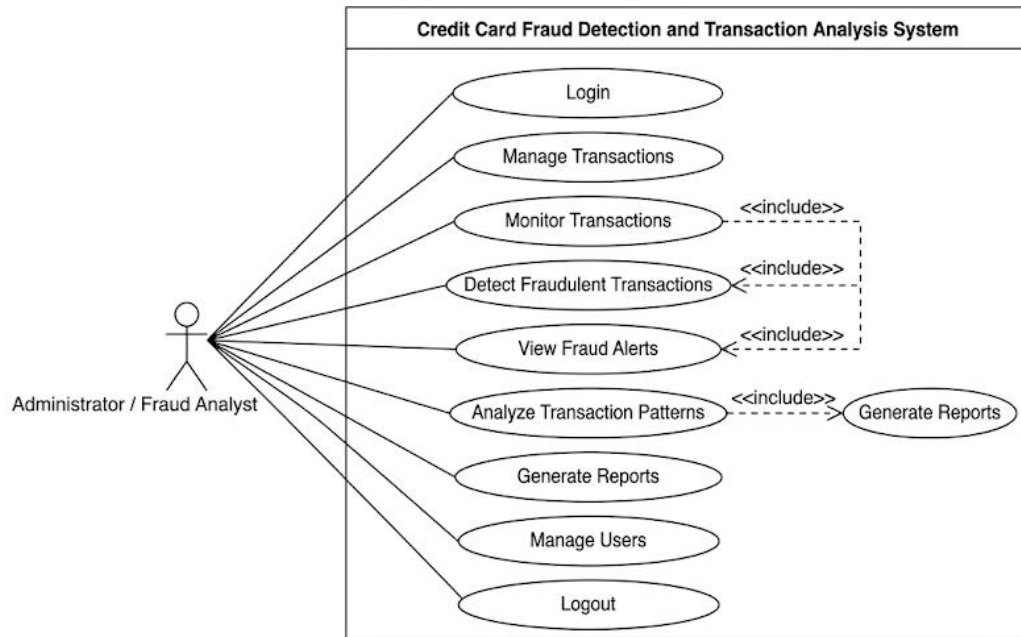


Figure 3.1: Use Case Diagram

This Use Case Diagram will depict user-level interactions with the proposed financial transaction monitoring and fraud detection system at a functional level. It will identify two main actors, the Administrator and the Financial Analyst, along with the specific system functionalities that would be available to each user role. The diagram will show how users authenticate, monitor transactions, analyze fraud alerts, manage fraud detection rules, and generate reports, thus giving an overall view of the system's functional requirements at a high level.

Class Diagram

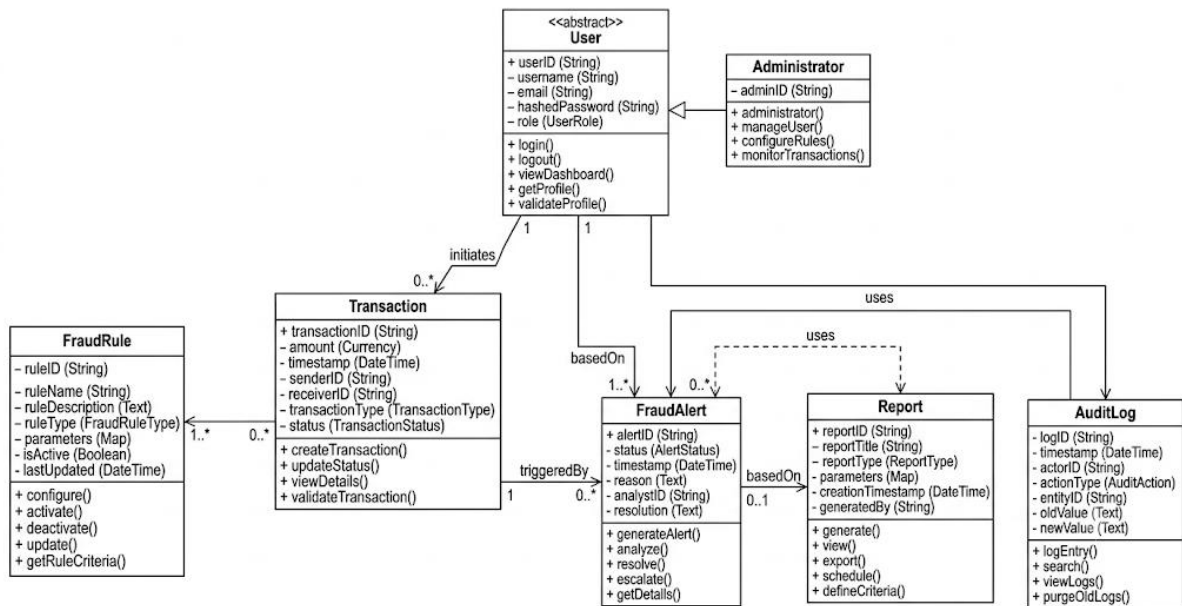


Figure 3.2: Class Diagram

This Class Diagram aims at illustrating the system's static structure. It starts by portraying the major classes, their properties, methods, and relationships. Classes such as User, Transaction, FraudRule, FraudAlert, Report, AuthenticationService, TransactionService, and FraudDetectionService will be displayed to indicate the structure of data and functionalities of the system. The relationships among these classes would guide the system development with the help of React.js, Python, and MySQL.

Activity Diagram

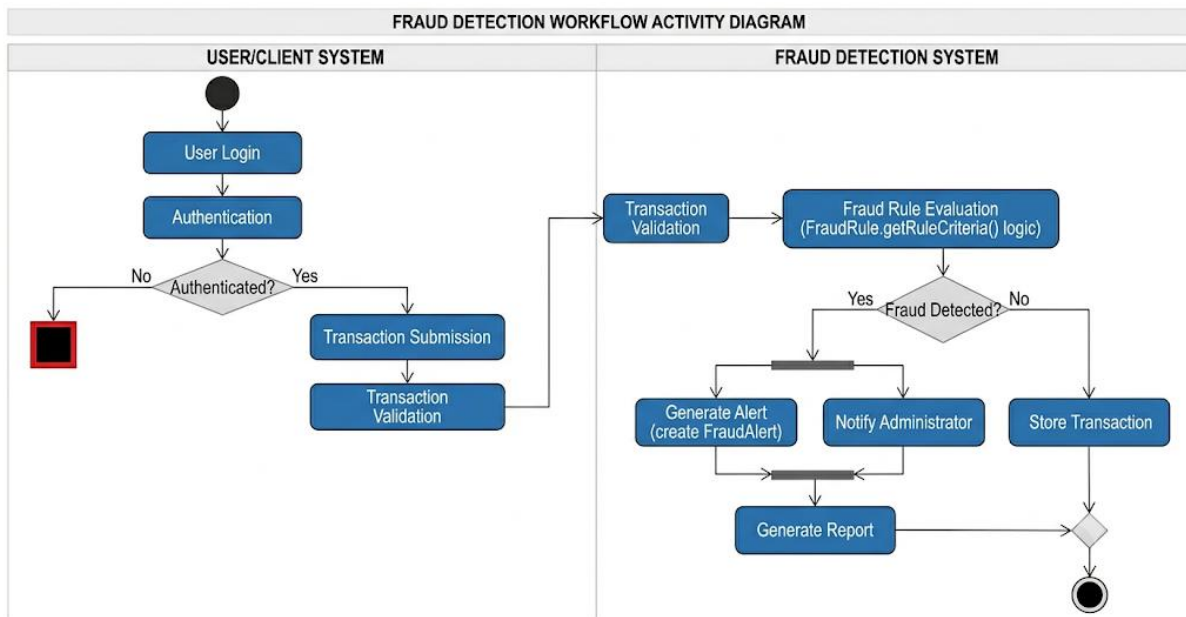


Figure 3.3: Activity Diagram

This Activity Diagram will depict the system workflow when processing financial transactions. User authentication will be the first step in the sequence of activities followed by transaction submission, validation and fraud rule evaluation. Decision elements will decide if the transaction is legitimate or suspicious after which the respective measures such as transaction storage, fraud alert generation and administrator notification will be carried out. Hence, the diagram will depict the fraud detection process flow clearly.

3.2.4 Architectural Design

The designed system will be based on a 3-tier architecture: presentation tier, application tier and database tier. This architecture shall be chosen due to the benefits of scalability, maintainability and separation of responsibilities. Every layer will have specific functions, and communication will be enabled by secure application programming interfaces (APIs).

The presentation layer shall present the graphical user interfaces via which users shall be able to access the system via a web browser. This layer shall be presented with user authentication, dashboard visualisation, transaction monitoring and report generation interfaces. All data input by the users will be passed to the application layer for processing.

The business logic for transaction validation, fraud detection, user management, report generation and notification services shall be located at the application layer. Rules to detect fraud shall be applied in this layer before the outcome of the transactions shall be

returned to the presentation layer. This middle layer shall act as the coordinator of communication between the presentation layer and the database layer.

The database layer shall store information about the system, user accounts, transaction records, fraud alerts, fraud rules and audit logs. Data integrity shall be ensured by use of relational database constraints, and secure data retrieval shall be made when requests are received from the application layer.

The three layers will communicate via secure HTTP requests and structured SQL queries. Any changes to a layer shall have minimal impact on the other layers so as to make the system flexible and easy to scale up in the future.

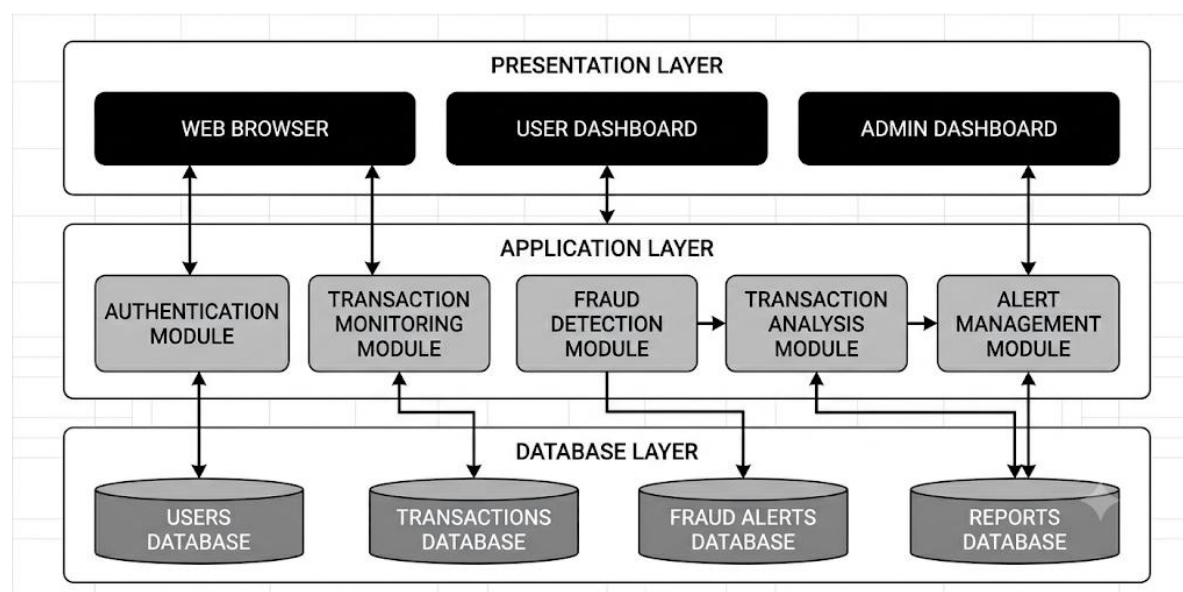


Figure 3.4: Three-Tier System Architecture

3.2.5 Database Design

The logical structure by which the data in the system shall be organised, stored and retrieved shall be defined by the database design. An efficient transaction record database shall be established using a relational database management system that provides consistency, integrity, and efficient querying of the database. The database shall have the ability to meet user authentication, transaction management, fraud analysis, and report generation.

Entity Relationship Diagram (ERD)

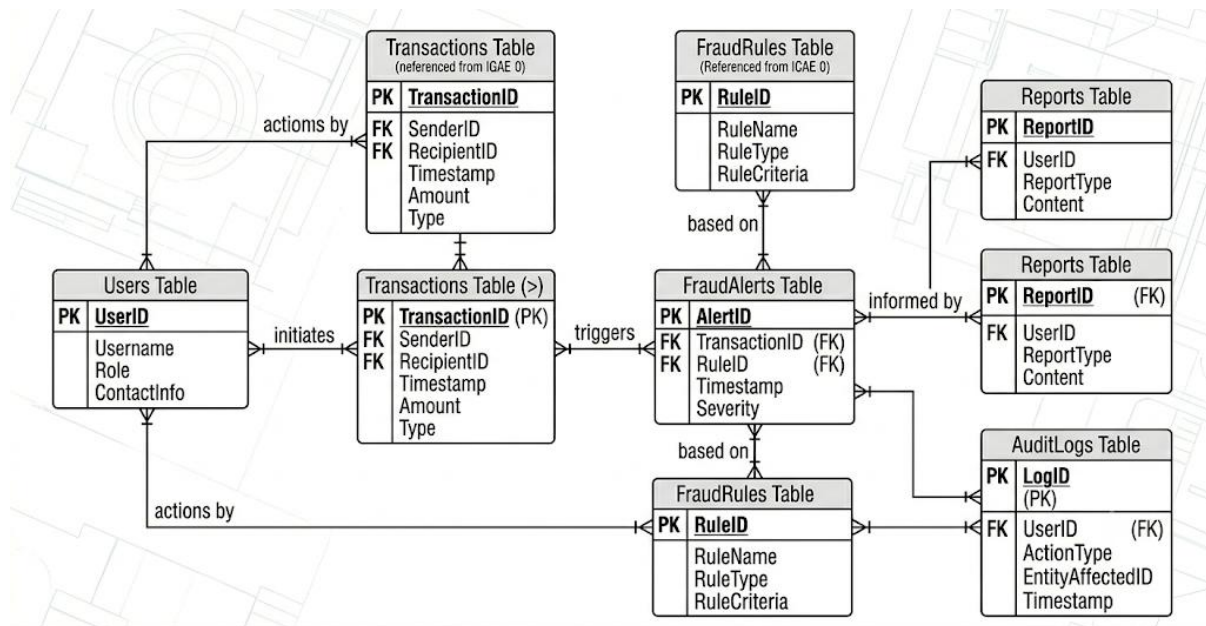


Figure 3.11: Entity Relationship Diagram

Database Schema

Table	Fields	Data Type	Key
Users	user_id, full_name, email, password, role	INT, VARCHAR, VARCHAR, VARCHAR	Primary Key
Transactions	transaction_id, user_id, amount, transaction_type, transaction_date, status	INT, INT, DECIMAL, VARCHAR, DATETIME, VARCHAR	Primary Key, Foreign Key
FraudRules	rule_id, rule_name, threshold, description	INT, VARCHAR, DECIMAL, TEXT	Primary Key
FraudAlerts	alert_id, transaction_id, rule_id, alert_status, detection_time	INT, INT, INT, VARCHAR, DATETIME	Primary Key, Foreign Keys
Reports	report_id, generated_by, report_type, generated_date	INT, INT, VARCHAR, DATETIME	Primary Key, Foreign Key
AuditLogs	log_id, user_id, activity, activity_date	INT, INT, TEXT, DATETIME	Primary Key, Foreign Key

The database should be normalised to Third Normal Form (3NF). Therefore, there should be a minimum level of data redundancy so that each attribute which is not a key attribute is

dependent on the primary key attribute. Functional dependencies will be removed by creating logically independent tables. As such, there shall be significantly fewer update anomalies, insertion anomalies and deletion anomalies in the system, and data integrity shall be preserved during system operation.

3.3 Computational Framework

The computational framework shall outline the logical processes by which fraud detection shall be accomplished. All financial transactions will be subjected to a rule-based algorithm which will be used to assess each transaction against pre-defined fraud detection rules.

Algorithm: Rule-Based Fraud Detection

Pseudocode

BEGIN

Authenticate User

IF login is successful THEN

 Receive Transaction

 Validate Transaction Data

 FOR each Fraud Rule

 IF transaction satisfies fraud condition THEN

 Generate Fraud Alert

 Store Alert

 Notify Administrator

 ELSE

 Store Transaction

 ENDIF

 ENDFOR

ELSE

 Display Authentication Error

ENDIF

END

The computational complexity of the fraud detection algorithm should be $O(n)$, where n shall be the number of fraud detection rules that are to be evaluated on each transaction. Each transaction will be evaluated against each rule once, so the execution time will be linear to the number of rules. This computational technique should then perform sufficiently in small- and medium-sized environments of financial transactions.

3.4 Tools

The implementation of the proposed system shall be supported by appropriate hardware and software tools selected to ensure reliable development, testing and deployment.

Hardware Requirements

Component	Specification
Processor	Intel Core i5 or AMD Ryzen 5
RAM	8 GB Minimum
Storage	256 GB HDD Minimum
Internet	Broadband Connection

Software Stack

Component	Tool
Operating System	Windows 11
Frontend Framework	React.js
Backend Language	Python 3.12
Backend Framework	Fastapi
Database	PostgreSQL
API Testing	Postman
IDE	Visual Studio Code
Version Control	Git and GitHub

Browser	Google Chrome
Package Manager	npm
Python Environment	Virtual Environment (venv)

Libraries and Frameworks

Frontend

- React.js
- React Router
- Axios
- Bootstrap 5
- Chart.js

Backend

- Flask-CORS
- SQLAlchemy
- PyMySQL
- Pandas
- NumPy

Dataset

The system shall be tested with simulated financial transaction data of both authentic and fake transactions. Test data will contain data for transaction ID, user accounts, transaction amounts, timestamp, payment methods and fraud status indicators. These datasets will be used to test the proposed rules used to detect the frauds.

Evaluation Platform

System tests will be carried out with a local development environment set up with XAMPP. Pre-deployment testing shall include functional testing, usability testing, performance testing and database integrity testing to ensure the system requirements are met.

CHAPTER FOUR

RESULTS AND DISCUSSION

4.1 Overview of System Implementation

The results of the implementation of Vantage, a web-based corporate card monitoring and credit card fraud detection system, are provided in this chapter. The system offers a single point of transaction recording, monitoring, fraud alert management and transaction analysis. The results presented in this chapter are based on the available system screenshots and the functionalities demonstrated in the application.

The system interface consists of two parts: monitoring and organisational management. Each monitoring module is the dashboard, transaction recording, transaction history, alerts, analysis and reports. The organisational management modules comprise departments, cards, users, detection rules and audit logs. The modules can be accessed via the application's navigation panel, and the dashboard displays available transaction information.

The system allows simulating corporate card transactions and offers an interface for viewing transactions. The interface includes the following components: a fraud detection rules engine and alert management facilities for searching and filtering alerts. The system also offers a dashboard of transaction summary and departmental expenditure details.;

4.2 Presentation of System Outputs

The following shows examples of the outputs of the Vantage system using screenshots of the graphical user interfaces. Discussions on the output based on the functions exhibited in the respective interfaces.

4.2.1 Central System Dashboard Module

The central system dashboard offers a view of the transaction monitoring environment. The interface includes summary cards of total transactions, suspicious transactions, open alerts and active cards. This also consists of views on transactions activity over the last 14 days, new alerts generated, departmental spend and recent transactions.

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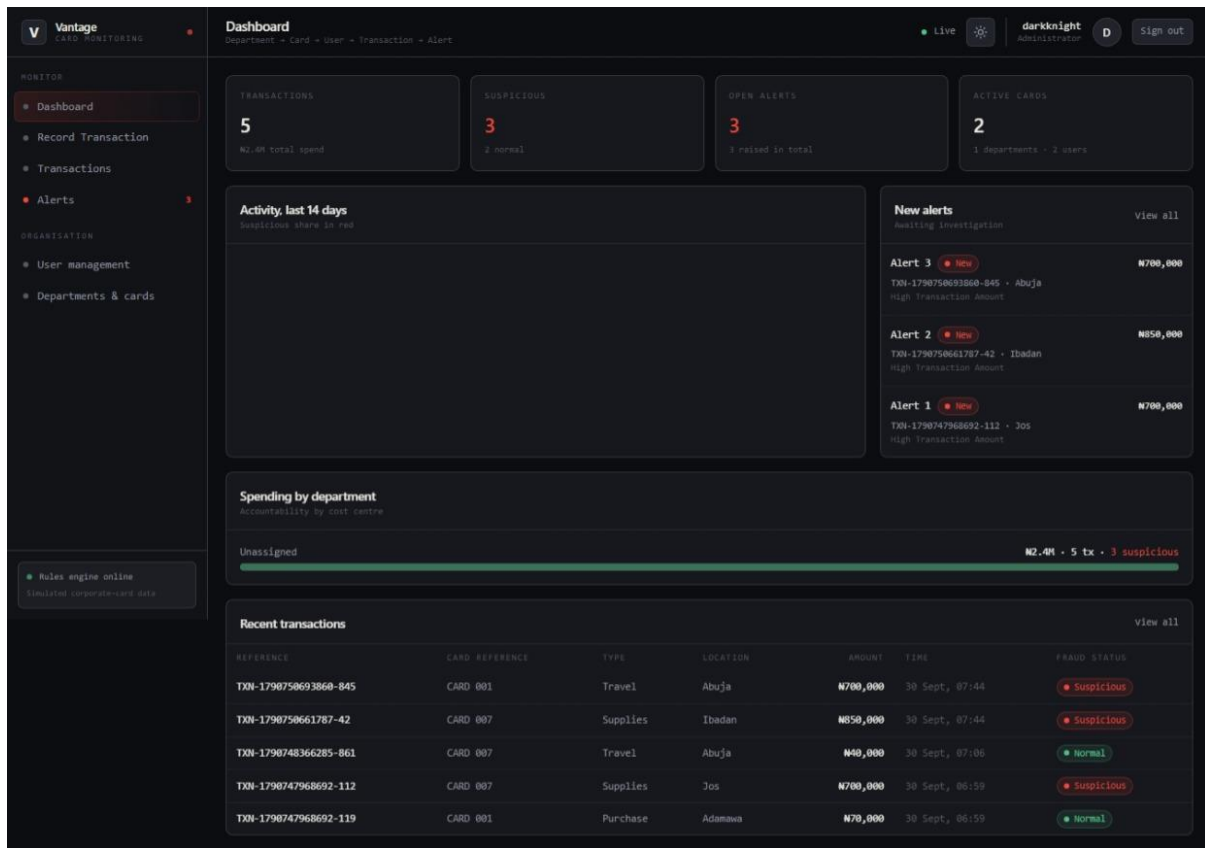


Figure 4.1: Central System Dashboard Interface

The dashboard screenshot shows one transaction that has been recorded; the total expenditure is ₦25,000. There are no suspicious transactions or open alerts shown, and the number of active cards is zero. The recent transaction section also displays details of the transaction such as reference, location, amount and status. The dashboard components present a single place to view transaction information and monitoring indicators

4.2.2 Transaction Recording and Simulation Engine Interface

The transaction recording interface includes fields to record the information that you need for a simulated corporate card transaction. The following information is shown: transaction reference, card reference, transaction amount, type of transaction, location, date and time. There is a submission control that is labelled Submit for evaluation to start the transaction evaluation process.

Figure 4.2: Transaction Entry and Simulation Interface

The image displays a form for entering a transaction with the details TXN-635734 for the transaction reference, 4417 for the masked card reference, ₦25,000 for the transaction amount, Online Purchase for the transaction type, and Lagos for the transaction location. There's also a transaction timestamp in the date and time field. The interface suggests that the transactions submitted are to be checked by the rules engine. The displayed screenshot does not reflect the result of the specific transaction submission, however.

The transaction history section has been included with the recording form. It has filters for reference, transaction type, location, fraud status, amount and date range. These controls provide a means of retrieving and examining recorded transaction information.

4.2.3 Transaction History and Monitoring Module

The transaction history has been incorporated in the transaction recording interface, and it contains filtering controls to view transaction records. The controls that can be used are transaction reference, transaction type, location, fraud status, minimum transaction amount, maximum transaction amount, and date range.

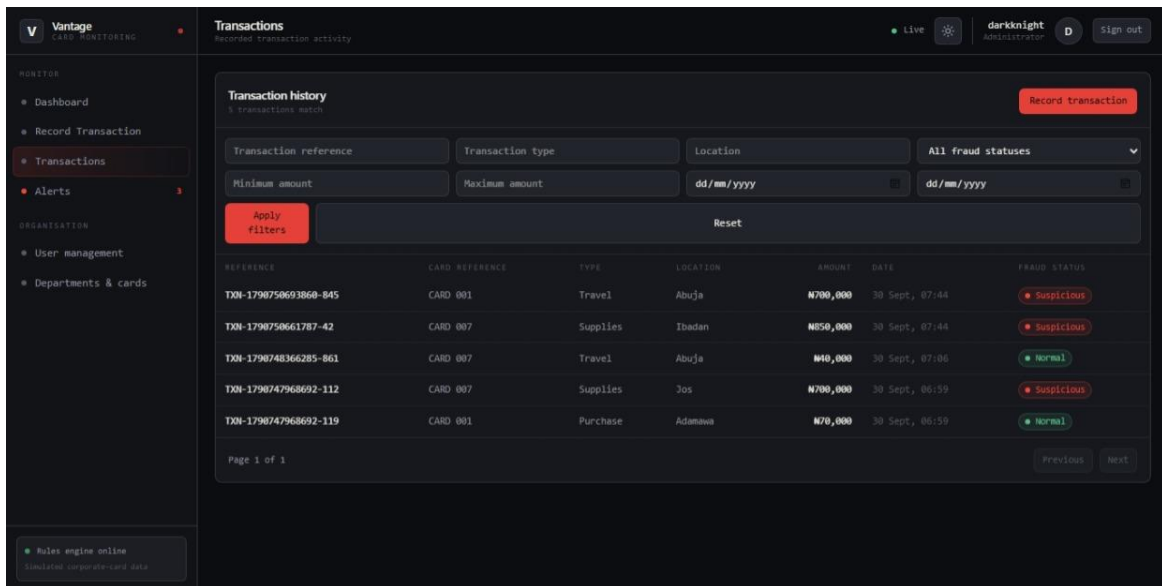


Figure 4.3: Historical Transaction Monitoring Interface

The interface should facilitate the retrieval and examination of the transactions recorded through different search criteria. However, the transaction history panel on the available screenshot doesn't show a list of completed transactions but rather a loading message. Thus, none of the filtering options can be rated as successful, based on the available screenshot.

4.2.4 Fraud Alert Management Queue

If a transaction crosses pre-defined transaction and/or risk limits (e.g., single-transaction amount thresholds), an alert is created in the fraud investigation queue. Details of the rule break, review states changes, or alerts can be viewed by analysts and even resolved.

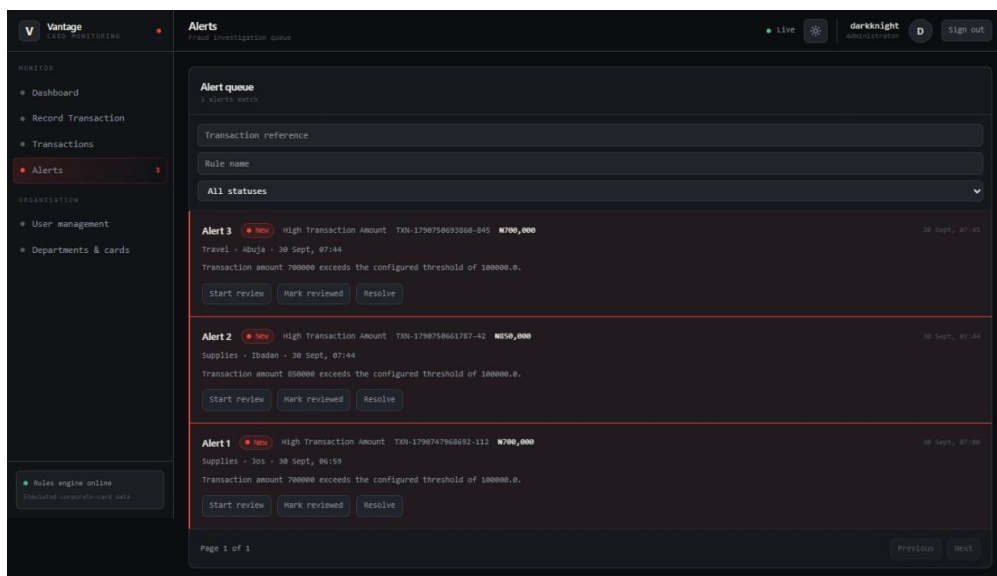


Figure 4.4: Fraud Alert Investigation Queue

The fraud alert management interface offers a main queue to review suspicious transaction alerts. The visible screen will have search boxes for transaction numbers, user and card. Additional filtering controls are provided for alert status, severity and department.

The screenshot shows there are no alerts in the queue and that no alerts are matching the current view. Thus, the interface shows users that an alert management facility is available and that there are search and filtering controls available. The screenshot, however, is not representative of the generation of a fraud alert, review or resolution.

4.2.6 User Access Control and Role Management Module

User management module is used to manage user provisioning and role-based authentication. Administrators can create new accounts, assign roles (Administrator / Analyst) and control user access to profiles in the platform.

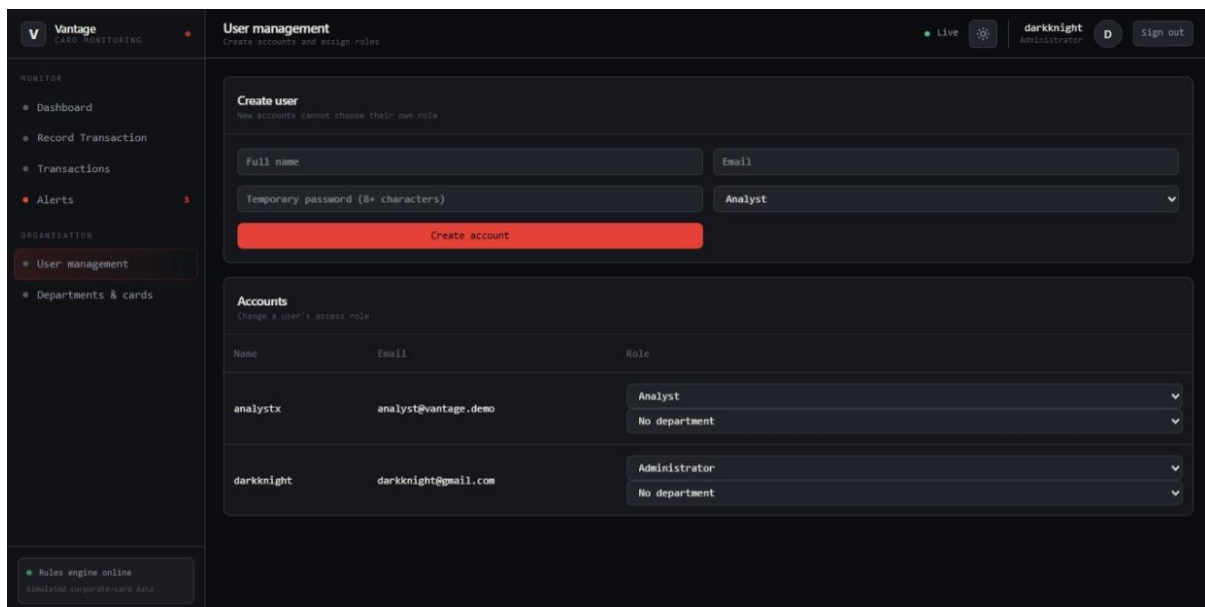


Figure 4.5: User Management and Role Assignment Interface

The account administration interface is shown in figure 4.6. Administrators can type in the user's credentials and set which roles they have and what permissions they have on the system when they were registered.

4.2.5 User Authentication and Account Registration Module

Credential authentication is done through encrypted credentials in the Vantage administrative console. The sign-on screen provides access for authorised users, and the analyst registration screen provides access for new analysts to get their sign-on credentials.

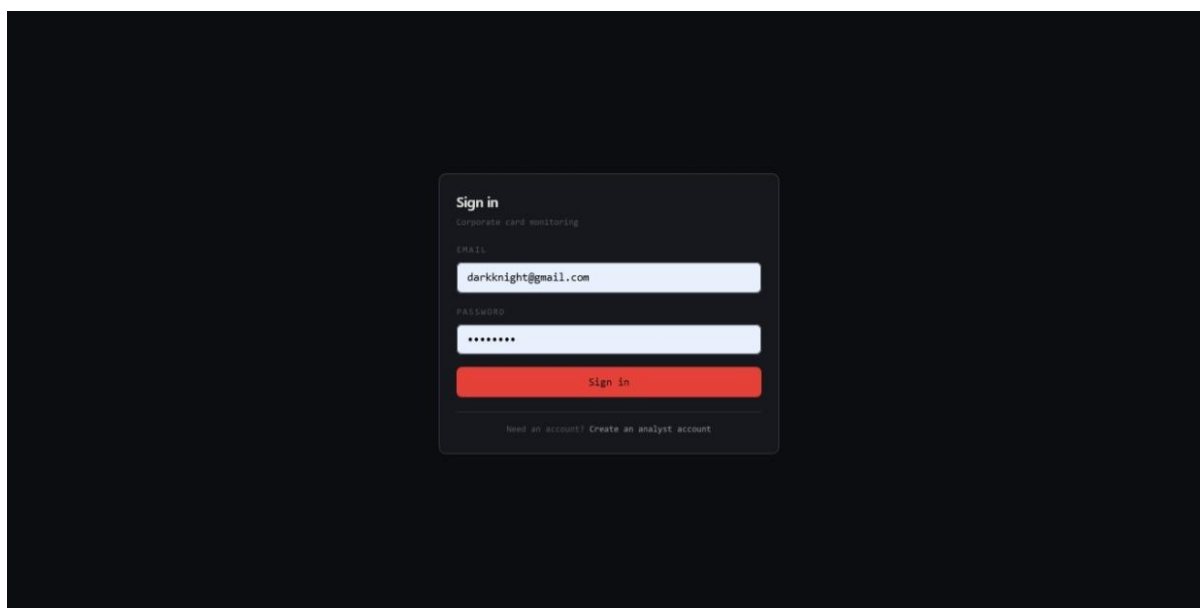


Figure 4.6a: User Sign-In Interface

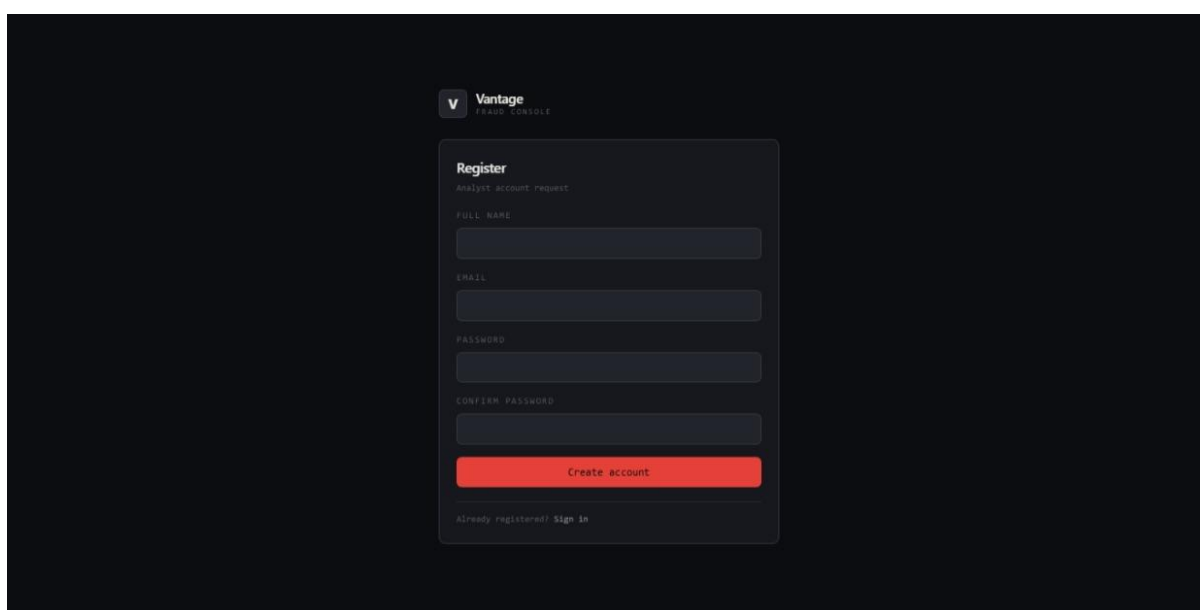


Figure 4.6b: Analyst Account Registration Interface

Figure 4.6a and Figure 4.6b show how to get into the web application. Figure 4.1b is the user registration page to be used for submitting a new analyst access request, and Figure 4.1a is the sign-in modal that appears when a user clicks the Sign In button.

4.2.6 Organisation Setup (Departments and Cards Management) Module

The Vantage application also offers navigation options for dealing with departments, cards and users. These options are presented under the organisational management section of the application. The detection rules and audit log interfaces are also available on the navigation panel.

These modules are part of the system's organisational and monitoring system. When images showing their functions are available, the way they actually operate and their outputs will be discussed.

The screenshot displays the 'Organisation setup' interface within the Vantage application. The top navigation bar includes the Vantage logo, 'CARD MONITORING', a 'Live' status indicator, a settings icon, the user 'darkknight administrator', and a 'Sign out' button. The left sidebar lists navigation options: 'MONITOR' (Dashboard, Record Transaction, Transactions, Alerts) and 'ORGANISATION' (User management, Departments & cards). The main content area is titled 'Organisation setup' with the subtitle 'Create departments and issue cards'. It features two primary forms: 'Create department' and 'Create card'. The 'Create department' form includes fields for 'Department code (e.g. FIN)', 'Department name', and 'Description (optional)', with a red 'Create department' button. The 'Create card' form includes fields for 'Card reference (e.g. CARD-1001)', 'Masked number (e.g. **** * 1001)', 'Select department' (a dropdown menu), 'Leave unassigned' (a dropdown menu), 'Card type (optional)', 'Issue date (optional)' (dd/mm/yyyy), and 'Expiry date (optional)' (dd/mm/yyyy), with a red 'Create card' button. Below these forms, there are sections for 'Existing departments' and 'Existing cards'. The 'Existing departments' section shows a table with one entry: 'FIN - Technical Department' with 'No description - active', and buttons for 'Edit' and 'Deactivate'. The 'Existing cards' section shows a table with two entries: 'CARD 001 - ***** 1001 Department 1 - active' and 'CARD 007 - ***** 1007 Department 1 - active', each with 'Edit' and 'Deactivate' buttons. A status box at the bottom left indicates 'Rules engine online' and 'Simulated corporate-card data'.

Figure 4.7: Organisation Setup (Departments & Cards Management) Interface

4.3 Discussion of Findings

The results of the study are discussed in light of the four objectives of the study as discussed below. The discussion is limited to the functions that are shown by the outputs that are available of the system.

The first goal was to study the transaction monitoring for a web-based system used for credit card transactions. The outputs available indicate that Vantage offers a dashboard to display transaction summaries and a transaction recording interface to simulate corporate card transactions.

The transaction recording interface features fields for transaction reference, transaction reference for card, amounts, transaction type, location and date and time. The transaction history also includes search and filtering controls to view transaction history. The features indicate the capability of the application to record and monitor transactions.

The dashboard also includes summary indicators of the number of transactions, suspicious transactions, open alerts and active cards. There is one transaction that was displayed, which is equal to ₦25,000, and there are no suspicious transactions or open alerts displayed. The outputs from this are therefore an example of the presentation of transaction information and provision of monitoring facilities. The ongoing processing of transactions, however, and the time required for the monitoring mechanism to act have not been determined based on existing evidence.

CHAPTER FIVE

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

5.1 Summary of the Study

In Chapter One, the introduction to the study was presented, where credit card fraud detection and transaction analysis in electronic payment systems were discussed. The problem statement was that it was a challenge to identify transactions that had suspicious characteristics and to track the activities of transactions and analyse the abnormal spending patterns. The goal and objectives of the study were set to explore the transaction monitoring functions of a web-based system, review fraud detection methods, discuss transaction analysis functions and determine the value of web-based transaction analysis for assisting with the investigation of fraud. The chapter also included the research questions, the importance and scope of the research and the operational definition of terms.

Chapter Two examined the literature of web-based systems for fraud detection and transaction analysis. The review covered the existing transaction monitoring, rule-based and pattern-based fraud detection techniques, and transaction analysis techniques for identifying suspicious transactions. Empirical studies related to the credit card fraud detection and transaction monitoring were also reviewed to understand the existing approaches, findings and limitations. Based on the review, gaps were identified in areas of transaction monitoring, fraud detection, transaction analysis and investigative support in a web-based environment.

The methodology used in this study was presented in Chapter Three. The chapter discussed the conceptual and theoretical frameworks, the iterative waterfall software development methodology and the requirements engineering process. It also covered the functional and non-functional requirements, system design using Unified Modelling Language diagrams, the three-tier system architecture and the database design. Also, the computational methods for rule-based fraud detection and hardware and software tools to be used in the development and testing of the system were introduced. These methodological elements were used to build the Vantage web-based corporate card monitoring and fraud detection system.

The results and discussion of the Vantage system outputs were presented and discussed in Chapter Four. It explored the system's graphical user interfaces such as Central Dashboard, Transaction Recording & Simulation, Transaction History & Monitoring and Fraud Alert Management. Available outputs showed how to provide facilities to record simulated transactions, view transaction summaries and manage fraud alerts. The discussion was conducted in the context of the four aims of the study and concerned the following topics:

transaction monitoring; fraud detection techniques; transaction analysis; and the usefulness of web-based transaction analysis as an aid to fraud investigation. In understanding the capabilities of the system, the shortcomings in the evidence that was available to operate were identified as well.

The summary of the study, conclusions and recommendations are presented in Chapter Five. A summary of the important findings of the study in relation to the aim and objectives is presented in this chapter, which draws conclusions from the findings presented in Chapter Four. Guidance is offered on the use and further development of web-based transaction monitoring and fraud detection/transaction analysis systems. The chapter also points out aspects of further research, including the effectiveness of detection, the use of larger transaction sets and improvements to the investigative functions in corporate card monitoring systems.

5.2 Conclusion

The study finalised that the Vantage web-based system offered a centralised system for the recording, monitoring and handling of corporate card transactions and for the management of alerting for card fraud. This system included a dashboard to view transaction summaries, a transaction recording interface to execute simulated transactions and an alert management interface to organise suspicious transaction information. These features proved the viability of developing the transaction monitoring, fraud detection and transaction analysis as one web-based application. Moreover, the system featured tools for reviewing transaction history and issuing alerts, potentially aiding in the detection and investigation of fraudulent or unusual transactions. The results, however, were not sufficient to confidently evaluate the effectiveness of fraud detection techniques, nor the usefulness of the analytical features. Based on this result, it was decided to develop a functional system for the monitoring of corporate cards on the Web, but further tests with simulated transaction cases and measurable evaluation criteria are needed to determine the detection performance and overall effectiveness of the developed system.

5.3 Recommendations

1. Web-based transaction monitoring systems with centralised transaction recording, monitoring in real-time and transaction history access should be implemented in financial institutions and organisations that issue corporate cards to enhance monitoring of card activities.

2. Fraud detection systems must also be able to include rule-based and pattern-based detection methods that are configurable, and the criteria for detection must be reviewed regularly to improve detection of fraudulent transactions while reducing the number of false alarms.
3. Arrange organisations to use a web-based transaction analysis tool with transaction filtering and expenditure summary features, which include graphical dashboards, to help identify unusual spending patterns and for better analysis of transaction records.
4. Organisations should enhance their fraud investigation procedures and ensure that the transaction analysis capability is linked to alert management facilities, enabling the prioritisation of alerts, trace of transactions, maintenance of investigation records and reporting of the results of the investigation.

5.4 Suggestion for Further Studies

To build upon the findings of this study, the following areas are suggested for future research:

1. Hybrid Detection Engines: Combining rule-based logic with unsupervised machine learning models (such as Isolation Forests or Autoencoders) to identify complex behavioral anomalies alongside policy threshold breaches.
2. Dynamic Threshold Tuning: Developing automated algorithms that dynamically adjust fraud rule thresholds based on historical cardholder spending patterns rather than fixed static limits.
3. Large-Scale Load Performance Testing: Evaluating system latency, database index optimizations, and concurrent message queue throughput under large-scale streaming datasets exceeding millions of transactions.

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